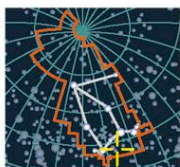


EMISSION NEBULA

IC 1396

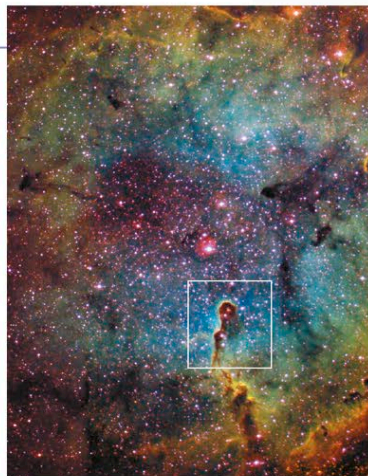


CATALOG NUMBER
IC 1396

DISTANCE FROM SUN
3,000 light-years

CEPHEUS

Occupying an area hundreds of light-years across, the IC 1396 complex contains one of the largest emission nebulae close enough to be observed in detail. It has an apparent diameter in the night sky 10 times that of a full moon. The mass of the nebula is estimated to be an immense 12,000 times the mass of the Sun, mainly consisting of hydrogen and helium in various forms. HD 206267, a massive, young blue star at the center of the region, produces most of the radiation that illuminates the nebula's interstellar material. Observations have shown that ionized clouds form a rough ring around this star at distances between 80 and 130 light-years. These clouds are the remains of the molecular cloud that originally gave birth to HD 206267 and its siblings, which compose the star cluster known as Tr37. Tracts of cool, dark material lie farther away from HD 206267. Among the most dramatic of these



GIGANTIC STELLAR NURSERY

The immense IC 1396 complex of emission nebulae, dark nebulae, and a young star cluster is shown here in a composite image. Mu Cephei is located at the center, and the Elephant's Trunk Nebula is boxed.

structures is one commonly known as the Elephant's Trunk Nebula. Research suggests that some of this material has been blown away from the star by strong stellar winds, causing the material to form elongated structures such as the Elephant's Trunk. Some of these structures stretch radially away from HD 206267 for up to 20 light-years. Within IC 1396 lies Mu (μ) Cephei, also known as Herschel's Garnet Star. One of the largest and brightest stars known, Mu Cephei is a red supergiant emitting 350,000 times the power of the Sun.

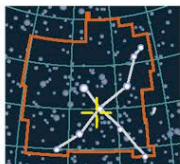
ELEPHANT'S TRUNK

The Elephant's Trunk Nebula is sculpted from a huge cloud of interstellar material in which star formation may take place in the future.



EMISSION NEBULA

DR6

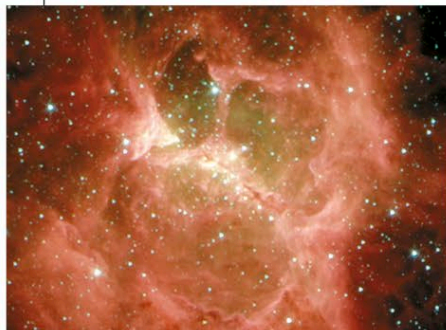


CATALOG NUMBER
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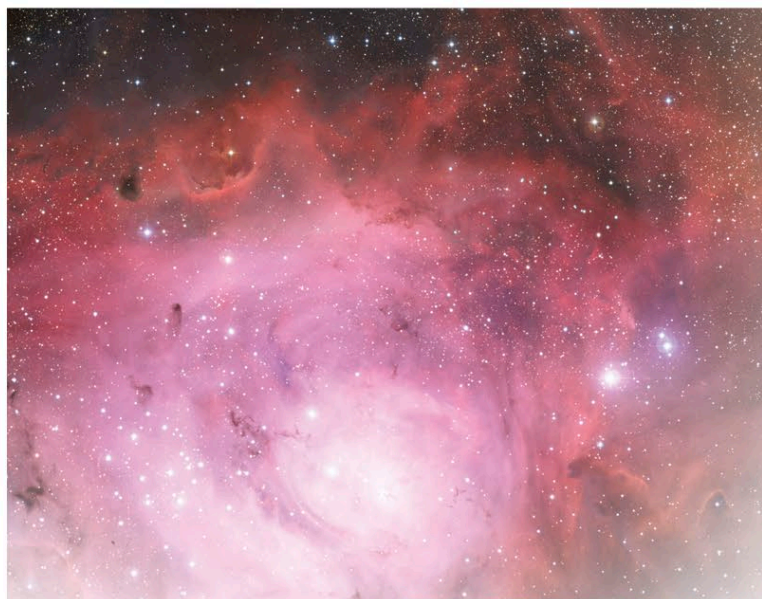
DISTANCE FROM SUN
4,000 light-years

CYGNUS

Strong stellar winds from about 10 young stars at the center of this unusual nebula have created cavities within its interstellar material, making it resemble a human skull. The nebula has a diameter of about 15 light-years, and the "nose," where the stars that have sculpted the nebula are located, is about 3.5 light-years across. The central group of stars is very young, having formed less than 100,000 years ago. The picture below is a composite of four infrared images.



GALACTIC GHOUL

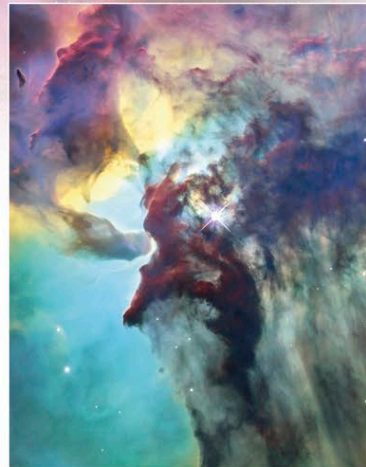


TWISTS OF GAS

Creative chaos is revealed within the vast Lagoon Nebula, as radiation and strong winds from newborn stars interact with surrounding clouds of interstellar dust and gas.

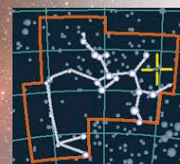
DARK GLOBULES

A Hubble Space Telescope image shows knots of dense gas and dust shaped by stellar winds from the nebula's newborn stars. Curtainlike streaks of dark and light show where the nebula is being eroded by the pressure of radiation.



EMISSION NEBULA

Lagoon Nebula



CATALOG NUMBERS
M8, NGC 6523

DISTANCE FROM SUN
5,200 light-years

MAGNITUDE 6

SAGITTARIUS

The Lagoon Nebula is a productive star-forming region situated within rich, conspicuous fields of interstellar matter. Covering an apparent diameter of more than three full moons, the Lagoon Nebula is so large and luminous that it is visible to the naked eye. The region contains young star clusters, distinctive Bok globules (see p.238), and very energetic star-forming regions. There are also many examples of twisted-rope structures thought to have been created by hot stellar winds colliding with cooler dust clouds. The bright center of the Lagoon Nebula is illuminated by the energy of several very hot young stars, including the 6th-magnitude 9 Sagittarii and the 9th-magnitude Herschel 36. Also found in the brightest region is the famous Hourglass Nebula (see p.263). The open cluster NGC 6530 (to the left of center in the main image) contains 50 to 100 stars that are only a few million years old. Clearly visible across the Lagoon Nebula are dark Bok globules.